

Madhav Gupta

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SKILLS

- Programming Languages & Databases: Python, Java, JavaScript (Node.js), C, SQL, MongoDB, PostgreSQL.
- DevOps & Cloud Security: GitHub Actions (CI/CD), Docker (Multi-stage), Terraform (IaC), AWS (VPC/EKS), Linux (Ubuntu/Kali).
- Security Tools & Analysis: Splunk Enterprise, Wireshark, Trivy (SCA), Checkov, SIEM Log Analysis, MITRE ATT&CK, UTM.
- Web & Frameworks: React.js, Node.js, RESTful APIs, Microservices.

EXPERIENCE

Manras Technologies

Zirakpur, India

Frontend Development Intern

June 2025 – Aug 2025

- Provided end-to-end technical support, completing **100%** of documentation, software updates, and troubleshooting requests within deadlines.
- Optimized Dynamics data models by refining record types, page layouts, and validation rules, reducing form errors by **20%** and improving system reliability.
- Created and managed **60+ high-priority** Visualforce and PDF documents for Adani Group real estate projects, ensuring accurate delivery of client-facing materials.
- Collaborated with team members on document assignments, streamlining handoffs and improving task completion efficiency across the portfolio.

Stony Brook University – AMS 161: Calculus II

Stony Brook, New York

Teaching Assistant

August 2025 – December 2025

- Assisted professor in facilitating coursework for **200+ undergraduate students**, reinforcing concepts in integration techniques, sequences, and series.
- Led weekly recitation sessions and office hours, clarifying problem-solving strategies and boosting student comprehension.
- Graded homework and exams with **100% accuracy** and timely feedback, ensuring fairness and consistency.

PROJECTS

SOC Detection Lab

Stony Brook, New York

- Designed a local security operations environment utilizing UTM virtualization to safely isolate attacker (Kali Linux) and victim (Ubuntu) network traffic.
- Deployed a Splunk Enterprise SIEM data pipeline, configuring Universal Forwarders to parse and index local system authentication logs (/var/log/auth.log) from a headless server.
- Conducted simulated network reconnaissance to capture .pcap telemetry in Wireshark, mapping the offensive behavior to the MITRE ATT&CK framework.

End-to-End DevSecOps Cloud Pipeline

Stony Brook, New York

- Provisioned secure AWS infrastructure (VPC, EKS) using Terraform and integrated Checkov to automatically scan IaC manifests, preventing cloud misconfigurations prior to deployment.
- Containerized a 3-tier web application (React, Node.js, PostgreSQL) utilizing multi-stage Docker builds and enforced non-root execution privileges to minimize the container attack surface.
- Engineered an automated CI/CD pipeline via GitHub Actions, integrating Trivy and npm audit for Software Composition Analysis (SCA) to proactively block critical CVEs from reaching production.

Astra

Stony Brook, New York

- Built a full end-to-end interpreter for a custom block-structured, ML-style language using Python and PLY (Lex/Yacc), covering lexing, parsing, AST construction, and execution.
- Designed a grammar-driven parser that converts source code into an Abstract Syntax Tree (AST) and enforces syntactic correctness via clear SYNTAX ERROR handling.
- Implemented AST-based semantic evaluation, embedding execution logic within node classes to support expressions, control flow, functions, and data structures.

FormFlow | Runner-Up (MongoDB Track), Code a Site Hackathon 2026

Stony Brook, New York

- Engineered a real-time, browser-based AI personal trainer utilizing a 17-point MediaPipe joint mesh to process raw webcam video into actionable biomechanical coaching, tracking joint telemetry and movement tempo at 60fps.
- Secured Runner-Up in the MongoDB track by developing a gamified backend that tracks rep-by-rep movement fluidity and instantly broadcasts form faults (e.g., knee valgus) to synchronized peer groups for gym accountability.

EDUCATION

Stony Brook University

Stony Brook, New York

B.S. in Computer Science

Expected Graduation, May 2028

- **GPA:** 3.66/4.00, Dean's List
- **Related Coursework:** Data Structures & Algorithms, Objects & Design, Computer Organization & Programming, Compiler Design, Database Systems, Object-Oriented Programming, Statistics & Applications